SIMATSENGINEERING(SAVEETHASCHOOLOFENGINEERING)

DepartmentofComputerScienceandEngineering

ASSESSMENT TOOL1–ANALYTICALPROBLEMSOLVING

CourseCode:	CSA0309	**CourseTitle:**	DataStructures
CO Assessed:	CO2–Manipulation(BL3,BL4) 30%	**Assessment:**	AssessmentTool1–Analytical ProblemSolving
Weightage:CO2	of CIA	**Total Marks:**	100
Statement:	Implementlineardatastructuressuchasstacks,queues,andlinkedlists,andanalyzethe e□ciencyoflinearandbinarysearchtechniques.		
StudentName:	SANKAR R	**Reg.No.:**	192525131
Section/Batch:	SLOT-B	**Date:**	17/07/2026

CodeofConduct

Iam SANKAR R(192525131)Name/RegNo)certifythatthissubmissionismyoriginalworkandthatIhave adheredtotheguidelinesspecifiedforthisassessment.Iunderstandthatanyviolationofacademicintegrityrul willresultindisciplinaryaction.

SignatureoftheStudent:SANKAR R

Instructions

- Thistestcontains80Analyticalproblem-solvingquestions.Total:100Marks.
- WhenastudentanswerusinganAnalyticalproblem-soiving,inevaluationtheanswershouldbewith followingfiverequirements:Understandingofproblem,selectappropriatedatastructure,Design the solution,analyze,Clarityofrepresentation.
- Eachstudentshouldwrite2questions.Nonegativemarking.Unansweredquestionscarryzeromarks.
- NoDTPprintoutsorelectronicdevicespermitted.Theyshouldprovidewrittenmaterials.

Section/Topic	Focus	QNos	Marks
LinkedDataStructure,SearchDataStructure	List Operations,BinarySearch	1	50
StackDataStructure,QueueDataStructures	StackNotations,PriorityQueue	2	50
	GRANDTOTAL:	100Marks	

3 | 5 | 7 | 10 | 15 | 20

Rubric3forCalculation

Criteria	Weightage	Level4–Exemplary	Level3–Proficient	Level 2–Developing	Level1–Beginning
ProblemUnder3tanding	20	Demon3trate3complete and preci3e under3tanding; identifie3allcon3traint3 andedge ca3e3	Show3clear under3tandingwith minor gap3	Partial under3tanding;mi33e33ome con3traint3	Mi3interpret3 problemorlack3 clarity
DataStructureSelection	20	Select3optimaldata 3tructurewith3trong ju3tificationand compari3on	Select3appropriate 3tructurewith rea3onableju3tification	Select33omewhat 3uitable3tructure withweakrea3oning	Incorrectornodata 3tructure3election
Algorithm De3ign	20	De3ign3 e□cient, logically3ound,and well-3tructured algorithm	Algorithmi3correct withminor ine□ciencie3	Algorithmi3partially correct or lack3 clarity	Algorithmi3 incorrectormi33ing
Accuracyof Solution	30	Error-free,well-3tructured,optimized code/3olution	Minorerror3but overallfunctional	Multipleerror3; partiallyworking	Non-functionalor mi33ing3olution
Clarity	10	Exceptionallyclear, well-organized,u3e3 diagram3effectively	Clearand organized withminor i33ue3	Somewhatorganized butlack3clarity	Poorlyorganized andunclear

FacultyIn-charge	Cour3eCoordinator	ProgramDirector
Signature: ____________	Signature: ____________	Signature: ____________
Date: ____________	Date: ____________	Date: ____________

Circular Queue (MAXSIZE = 6)

Step	Operation	Queue (Front – Rear)	Front	Rear	count
a.)	Initially empty		0	-1	0
b.)	Enqueue A, B, C	A | B | C | – | – | –	0	2	3
c.)	Dequeue A	– | B | C | – | – | –	1	2	2
D.)	Enqueue D, E, F	– | B | C | D | E | F	1	5	5
E)	Deque B, C	– | – | – | D | E | F	3	5	3
F.)	Enqueue G	G | – | – | D | E | F	3	0	4
G.)	Dequeue D, E	G | H | I | – | – | F	5	0	2
H.)	Enqueue H, I	G | H | I | – | – | F	5	2	4
I.)	Dequeue F	G | H | I | – | – | –	0	2	3
J.)	Dequeue G, H	– | – | I | – | – | –	3	2	1
K.)	Deque I	– | – | – | – | – | –	2	1	0

Final queue is empty.

(At step h) Index:

0	1	2	3	4	5
G	H	I	–	–	F

↑ Rear (index 0) ↑ Front (index 5)

Final: Front = 2

Rear = 1

Count = 0

2. Linked list configuration:

a.) L_1 append (10); L_1 append (20); L_1 append (15);

10 → 20 → 15 → Null

↑ current Position.

b.) L_2 append(10); L_2 append (20); L_2 append (15);

L_2 move To start ( ); L_2 insert (39); L_2 next ( ); L_2 insert(12);

10 → 39 → 12 → 20 → 15 → Null

↑ current Position.